

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
data = pd.read_csv('student_info.csv')
data.head()
```

```
data.shape
```

```
data.isnull().sum()
```

```
plt.scatter(x = data.study_hours , y=data.student_marks)
plt.title("Student Data")
plt.xlabel("Student Study Hours")
plt.ylabel("Student Marks")
plt.show()
```

```
data.mean()
```

```
data = data.fillna(data.mean())
data.isnull().sum()
```

```
X = data.drop(columns = 'student_marks')
y = data.drop(columns = 'study_hours')
X.shape , y.shape
```

```
from sklearn.model_selection import train_test_split
```

```
X_train , X_test , y_train , y_test = train_test_split(X, y , random_state=51 , test_size=0.2)
X_train.shape , y_train.shape , X_test.shape , y_test.shape
```

```
y_test
```

```
X_train
```

```
from sklearn.linear_model import LinearRegression
```

```
lr = LinearRegression()
lr.fit(X_train , y_train)
```

```
lr.score(X_test , y_test)
```

```
print(lr.intercept_)
```

```
print(lr.coef_)
```

```
pred = lr.predict(X_test)
```

```
pred
```

```
pd.DataFrame(np.c_[X_test , y_test , pred] ,columns=[ 'Study hours' , 'Original Marks' , 'Predicted Marks'])
```

```
# Fine Tune Model
```

```
lr.score(X_test,y_test)
```

```
plt.scatter(X_train, y_train)
```

```
plt.plot(X_train ,lr.predict(X_train) , color='r')
```

```
import joblib
```

```
joblib.dump(lr , 'Student_Marks_Prediction_Model.pkl')
```

```
model = joblib.load('Student_Marks_Prediction_Model.pkl')
```

```
model.predict([[ 1 ]])[0][0]
```

```
data.describe()
```